Phase Dynamics as Bridge between Thought and Behavior

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Abstract

■ Ongoing thoughts play a critical role in modulating cognitive performance, with phenomena such as mind wandering consistently associated with decreased task accuracy and prolonged RTs. However, the neural mechanisms underlying the influence of thought dimensions on cognition and behavior remain unclear. To elucidate this, we used EEG to investigate how two key thought dimensions—deliberate control (deliberate vs. spontaneous thoughts) and task relatedness (on-task vs. off-task thoughts)—modulate RT during a simple RT task. Behavioral results showed that both on-task and more deliberate thoughts were associated with shorter RTs compared to off-task and more spontaneous thoughts. Neurodynamic analyses revealed that on-task and deliberate thoughts were characterized by prestimulus increases in both frequency sliding, reflecting faster phase-based neural speed, and sample entropy, reflecting higher neural uncertainty/flexibility. Both prestimulus frequency sliding and sample entropy were significantly related to the degree of poststimulus intertrial phase coherence, which, in turn, correlated with RT. This sequential relationship suggests that phase-based neural dynamics play a crucial role in mediating the relationship of thought with task-related behavior. Together, these findings suggest that phase-based neural dynamics could play a key modulatory role across the divide of prestimulus and poststimulus activity in shaping the influence of ongoing thoughts (deliberate control and task relatedness) on task execution and its related behavior (RT). ■

INTRODUCTION

Human thought is highly dynamic, continuously shifting and evolving over time (Hua et al., 2022; Rostami et al., 2022; Scalabrini et al., 2022; Smallwood, Bernhardt, et al., 2021; Smallwood, Turnbull, et al., 2021; Girn, Mills, Roseman, Carhart-Harris, & Christoff, 2020; Kucyi, 2018; Christoff, Irving, Fox, Spreng, & Andrews-Hanna, 2016; Vanhaudenhuyse et al., 2011). For example, when "mind wandering" occurs, thoughts seamlessly transition between various contents, whereby attention drifts away from the current task at hand toward unrelated streams of thought (Smallwood, Turnbull, et al., 2021; Franklin, Smallwood, & Schooler, 2011; Smallwood & Schooler, 2006). During such states, the mind operates with remarkable freedom and fluidity, reflecting the dynamic nature of thought (Girn et al., 2020; Christoff et al., 2016, 2018). These fluctuations in the task relatedness and spontaneity of our thoughts have been shown to influence task-related behaviors, such as RT (Alperin, Christoff, Mills, & Karalunas, 2021; Girn et al., 2020; Christoff et al., 2016; Barron, Riby, Greer, & Smallwood, 2011; Smallwood & Schooler, 2006). Specifically, when the mind drifts away from the task at hand, RT tends to increase, whereas maintaining task-focused thoughts is associated with shorter RTs (Christoff et al., 2016; Barron et al., 2011; Smallwood & Schooler, 2006). Despite these insights, the mechanistic pathway that links ongoing thought with poststimulus response behavior remains poorly understood, leaving a critical gap in our understanding of how thoughts shape behavior.

A key aspect in understanding this interplay is the multidimensional nature of thought. Traditionally, thought has been dissected into two dimensions—task relatedness (on-task vs. off-task) and deliberate control (deliberate vs. spontaneous; Christoff et al., 2016, 2018; Barron et al., 2011; Smallwood & Schooler, 2006). The "goal-directed hypothesis" posits that deliberate control and task relatedness are inherently coupled, whereby spontaneous thought represents a failure of executive control to maintain task-focused attention, thus conflating off-task processing with spontaneous thought generation (Groot, Csifcsák, Wientjes, Forstmann, & Mittner, 2022; Seli et al., 2018; Irving, 2016; Barron et al., 2011; Smallwood & Schooler, 2006). Although this framework predicts that such executive control failures manifest behaviorally as prolonged RTs and diminished task performance (Seli et al., 2018; Smallwood & Schooler, 2006), emerging empirical evidence challenges this unitary conceptualization. Specifically, recent findings demonstrate that deliberate control and task relatedness constitute dissociable cognitive dimensions, as evidenced by instances where thoughts can be simultaneously off-task and deliberately maintained (Mills, Raffaelli, Irving, Stan, & Christoff, 2018). Therefore, in this study, we divided thought into these two dimensions to investigate how thought interacts with neural dynamics and task-related behaviors (Figure 1). Here, we investigated this overall goal in three separate aims (Figure 1). To achieve these aims, we

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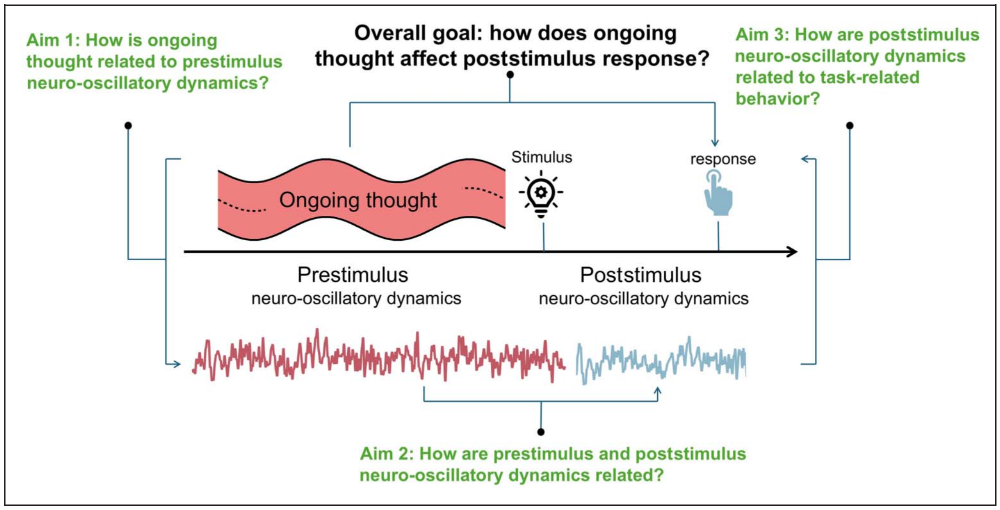

Figure 1. Conceptual framework of this study. Overall goal: To examine how ongoing thought influences poststimulus response through its interaction with prestimulus and poststimulus neural oscillations. Aim 1: To investigate how ongoing thought dimensions (on-task deliberate vs. off-task spontaneous) are related to prestimulus oscillations, including FS and SE. Aim 2: To determine how prestimulus oscillations (FS and SE) interact with poststimulus oscillations (ITPC). Aim 3: To explore how poststimulus oscillations, particularly ITPC, relate to behavioral responses (RT).

employed a simple RT task (RTT), categorizing ongoing thoughts based on task relatedness (off-task vs. on-task) and deliberate control (spontaneous vs. deliberate).

Thought and Prestimulus Dynamics (Aim 1)

The first aim was to examine how the two thought dimensions, task relatedness and deliberate control, affect neuro-oscillation dynamics. Most task-based and event-related studies focus on evoked brain responses during the poststimulus period (Barron et al., 2011; Smallwood & Schooler, 2006; Polich, 2003). However, it is important to also consider prestimulus with, for instance, its spontaneous brain oscillations that are known to impact poststimulus activity and its related behaviors (Northoff, Buccellato, & Zilio, 2025; Huang et al., 2017; Scalabrini et al., 2017; He, 2013). This is important for two reasons. First, there is strong evidence that thoughts are embedded and influenced by ongoing brain dynamics (Karapanagiotidis et al., 2018; Girn et al., 2017; Smallwood & Schooler, 2006). Second, prestimulus activity impacts poststimulus activity (Wainio-theberge, Wolff, & Northoff, 2020; Wolff et al., 2019; Huang et al., 2017; He, 2013), including the latter's associated mental features (Northoff et al., 2025; Wolff et al., 2019).

Therefore, in this study, we specifically focused on the impact of ongoing thoughts on prestimulus neural oscillatory dynamics. We focused on two dynamic features: neural speed and neural uncertainty. Neural speed can be measured using peak frequency sliding (FS), quantifying the rate of change in neural oscillatory activity—with higher FS indicating faster oscillations that reflect adaptive responses to cognitive demands (Hua et al., 2022; Cohen, 2014a, 2014b). Neural uncertainty, captured by sample entropy (SE), measures the unpredictability of oscillatory patterns, where higher SE suggests greater signal variability that may support cognitive flexibility (Lechner & Northoff, 2023; Lau, Pham, Chen, & Makowski, 2022; Liu, Ma, Fan, Abbod, & Shieh, 2018). We will examine how ongoing thought—integrating both task relatedness and deliberate control—relates to these prestimulus oscillatory metrics.

To operationalize these measures, we chose to focus on the alpha band (8–12 Hz) for several reasons. First, alpha oscillations are central to cognitive control and attention, serving as an input gating mechanism that regulates the flow of task-relevant information, for example, the input stream (Klimesch, 2012; Jensen & Mazaheri, 2010). Second, alpha oscillations have been shown to exhibit strong prestimulus dynamics that are known to predict poststimulus performance (Northoff et al., 2025; Foxe & Snyder, 2011; Palva & Palva, 2007). Finally, alpha band activity is sensitive to both thought and stimulus modulation (Northoff et al., 2025; Hua et al., 2022), making it an ideal frequency range for investigating the interplay between ongoing thoughts and neural preparation for subsequent external stimulus/input processing (Klimesch, 2012). These attributes make the alpha band a robust temporal window to make sound a priori hypotheses about the neural mechanisms underlying the impact of thoughts on behavior.

Most prior EEG studies of thought and cognition have emphasized stimulus-locked or whole-epoch activity (Barron et al., 2011; Smallwood & Schooler, 2006; Polich, 2003), thereby treating prestimulus and poststimulus periods as a single analytic window. However, this approach risks obscuring the distinct functional contributions of these two periods. The prestimulus interval reflects the spontaneous, ongoing neural dynamics that form the preparatory context in which thoughts unfold and external input is received (Northoff et al., 2025; Wainio-theberge et al., 2020; Huang et al., 2017; Scalabrini et al., 2017; He, 2013). In contrast, the poststimulus interval reflects stimulus-driven neural responses and their direct relationship to behavioral performance (Barron et al., 2011;

Smallwood & Schooler, 2006; Polich, 2003). By examining these periods separately, we are able to disentangle how ongoing thought shapes preparatory neural states before stimulus onset (Aim 1) and how these states subsequently influence stimulus-locked phase alignment and task execution (Aims 2 and 3). This methodological choice is therefore essential to reveal the mechanistic pathway from thought to behavior through neural dynamics.

Poststimulus Dynamics: Interaction with Prestimulus Dynamics (Aim 2) and Relation to Behavior (Aim 3)

Aims 2 and 3 address how prestimulus dynamics influence poststimulus neuro-oscillation dynamics (Aim 2) and behavior (Aim 3). In Aim 2, we explored how prestimulus FS and SE influence poststimulus intertrial phase coherence (ITPC), which quantifies entrainment (Lakatos et al., 2019) as temporal alignment of neural oscillatory phases to the timing of the external stimuli across trials (Lechner & Northoff, 2023; Northoff, Klar, Bein, & Safron, 2023; Rapela, Westerfield, & Townsend, 2018; van Diepen & Mazaheri, 2018). We related FS and SE in the prestimulus period to ITPC in the poststimulus period because we hypothesized that prestimulus neural oscillatory speed (indexed by FS) and uncertainty (indexed by SE) could influence the phase of neural oscillations at stimulus onset (indexed by ITPC). The poststimulus ITPC, in turn, provides the ground for the brain's neural activity to reach the optimal state required for behavior with faster RTs (van Diepen & Mazaheri, 2018). Such phase alignment subsequently affects the degree of neural oscillatory synchronization (Lau et al., 2022; Cohen, 2014a, 2014b), which shapes how the external input is processed including the participant's subsequent behavior, for example, RT. Given this, we hypothesize that enhanced prestimulus dynamics will lead to increased ITPC. In Aim 3, we investigate the behavioral consequence of this enhanced ITPC by testing its impact on RT. Given that higher ITPC is associated with more efficient stimulus processing and on-task thoughts (Duprez, Stokkermans, Drijvers, & Cohen, 2021; López, Pusil, Pereda, Maestú, & Barceló, 2019; Rapela et al., 2018; van Diepen & Mazaheri, 2018; Cohen, 2014a), we hypothesize that increased prestimulus FS and SE will be associated with higher poststimulus ITPC and that this will in turn lead to faster RT.

To formally test the sequential pathway hypothesized in Aims 1–3—namely, that ongoing thought influences prestimulus neural dynamics (FS, SE), which in turn modulate poststimulus ITPC and ultimately shape RTs—we employed various mediation model analyses. These models allowed us to go beyond pairwise associations and directly examine whether prestimulus FS and SE function as intermediaries between thought dimensions and poststimulus outcomes, as well as whether ITPC mediates the link between prestimulus dynamics and behavioral performance. Given that the central claim of this study concerns how thought impacts behavior through neural dynamics, mediation analyses provide a principled and necessary framework to establish this mechanistic pathway.

To provide a clear analytic roadmap, we followed three primary hypotheses corresponding to our three aims.

Aim 1 (from thought to prestimulus dynamics): We hypothesized that the two thought dimensions—task relatedness and deliberate control—would differentially influence prestimulus oscillatory dynamics. Specifically, on-task and deliberate thoughts were expected to be associated with higher FS and higher SE.

Aim 2 (from prestimulus to poststimulus activity): We hypothesized that enhanced prestimulus FS and SE would facilitate stronger phase-based temporal alignment at stimulus onset, as reflected in increased poststimulus ITPC.

Aim 3 (from poststimulus activity to behavior): We hypothesized that higher poststimulus ITPC would be associated with shorter RT, thereby linking thought-related prestimulus dynamics to task execution.

Beyond these three hypotheses, we conducted exploratory analyses to test the interdependence of FS and SE by examining their correlation during the prestimulus period. In addition, we carried out post hoc analyses to assess the robustness and specificity of the observed effects. These included (i) phase-shuffling controls to verify the phase dependence of FS, SE, and their ITPC relationships and (ii) resting-state analyses to evaluate whether the thought–neural dynamics associations were truly task related. This structure enables us to clearly distinguish a priori hypotheses from exploratory and post hoc extensions, while situating each analysis within the broader framework of the overall study goal.

Together, all three aims and hypotheses constitute a novel mechanistic pathway that links thoughts and ongoing neural dynamics (FS, SE) during the prestimulus period to poststimulus ITPC and task execution (RT; Nakatani, Kawasaki, Kitajo, & Yamaguchi, 2021; Hanslmayr et al., 2007). Specifically, this study reveals that on-task and deliberate thoughts are associated with higher prestimulus FS and SE, which correlate with greater poststimulus ITPC that by itself relates to shorter RT. These results underscore the influence of ongoing phase dynamics across the divide of prestimulus and poststimulus activity including their relation to task performance on thoughts. We provide new insights into how our thoughts and behavior are embedded in a highly influential neuro-oscillatory context of ongoing phase dynamics right at the interface of internal activity (prestimulus period) and external stimuli (poststimulus period).

METHODS

Participants

Forty right-handed adults participated in the study (13 women; age range = 18–33 years; mean age = 22.4 years, *SD* = 2.4 years). All had normal or corrected-to-normal vision and reported no neurological or psychiatric conditions that might affect performance (including, but not limited to, major depressive disorder, anxiety disorders, bipolar disorder, or schizophrenia). In addition, participants were instructed not to consume alcohol or engage in sleep deprivation on the day before the experiment. The experimental protocols were approved by the research ethics committee of the Zhejiang University and the University of Ottawa, and the study was carried out with their permission. Written informed consent was obtained from each participant before study participation.

The sample size ($n = 40$) was determined based on prior studies using EEG to investigate spontaneous thought and neural dynamics in similar paradigms (Hua et al., 2022; Girn et al., 2017; Barron et al., 2011), which typically included 25–40 participants. In addition, a priori power analysis was conducted using G*Power 3.1 (Faul, Erdfelder, Lang, & Buchner, 2007) to determine the required sample size for detecting medium effects (Cohen's $d = 0.5$) in paired-sample t tests, which constitute the primary statistical tests in this study. The analysis indicated that a minimum of 34 participants would be required to achieve 80% power at $\alpha = .05$ (two-tailed). Accordingly, we recruited 40 participants, which provides sufficient statistical power to detect the hypothesized effects.

Procedures

The procedure consisted of two sessions conducted with EEG recording: (1) a resting-state session and (2) a simple RTT, each lasting 10 min. To remain consistent with previous investigations, and because of the related time constraints (i.e., 10-min acquisitions), to ensure a longer prestimulus period (4 sec) to generate and assess ongoing thought (Rostami et al., 2022; Vanhaudenhuyse et al., 2011), both resting-state and task sessions incorporated 20 thought probes, during which participants responded to two questions (Figure 2A). Although the number of trials is not very large, ITPC analysis can still be conducted,

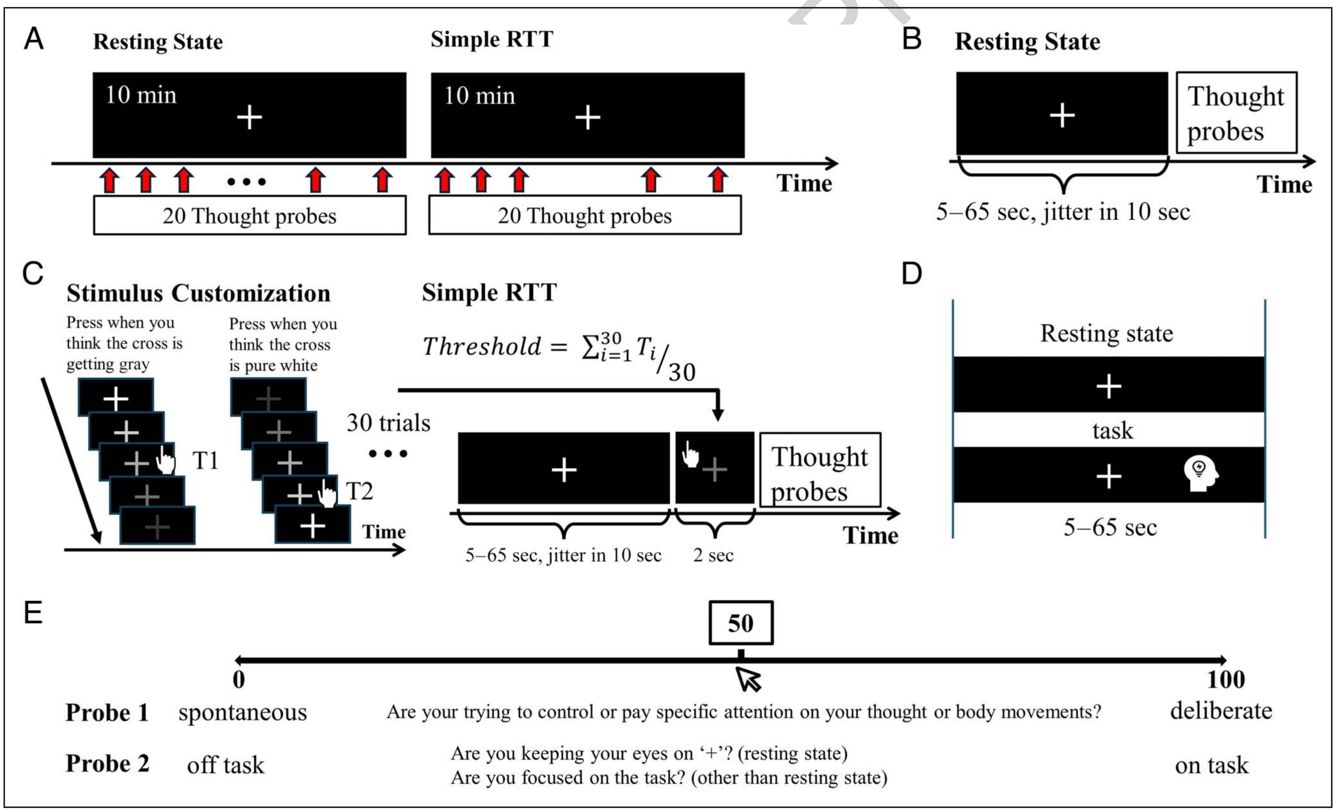

Figure 2. Experiment paradigm. (A) Overall experimental design. The procedure comprised two tasks: a resting state and a simple RTT, both of which lasted 10 min, where 20 thought probes were presented. (B) Resting state. Participants were instructed to focus on the fixation cross and remain relaxed. They were also prompted to respond to thought probes at random intervals (ranging from 5 to 65 sec, with jitters of 10 sec). (C) RTT. Before the task, participants' thresholds for identifying "white" and "gray" were determined. In the main part of the task, participants were asked to maintain their focus on the fixation cross, similar to the resting state, with one exception: Upon detecting the fixation cross turning gray, they were required to press the space bar with their left hand as quickly as possible. The fixation cross remained for 2 sec regardless of whether a response was made. The duration before the fixation cross changed was randomized between 5 and 65 sec with jitters of 10 sec. After the 2-sec response window, thought probes were presented. (D) Difference between preprobe period in resting state and prestimulus period in RTT. The IPI during resting state and ITI during RTT share the same time length, so the only difference is that in the RTT prestimulus period, participants need to prepare on both thought and body for the task response. (E) Thought probes. Participants were asked to evaluate the score of deliberate control and task relatedness using a visual analog scale ranging from 0 to 100.

as a stable result can be obtained with more than 15 trials (Cohen, 2014a).

The resting-state session was performed first, where participants were instructed to fixate on a cross displayed on the screen and remain relaxed until prompted by a thought probe. The interprobe interval (IPI) in the resting-state task varied from 5 to 65 sec, with a jitter of 10 sec (Figure 2B).

For the RTT, participants were required to respond upon detecting a color change. Significant interindividual variability in gray level difference judgments (Kuehni, 2009) was controlled for in this task (Figure 2C). Once the stimulus threshold was determined, the corresponding grey level was employed as the stimulus gray level in the RTT.

The intertrial interval (ITI) during the RTT was determined in the same way as the IPI during the resting state (Figure 2C), the only difference being that a behavioral response was required for each probe (Figure 2D). Thus, the resting state served as a control condition for task-preparatory states of thought. During the ITI, a fixation cross with a gray level of 100 (completely white) was presented. During the stimulus presentation period, the gray level transitioned gradually over 2000 msec from white to the threshold gray level. Participants were instructed to press the space bar as quickly as possible upon detecting the fixation cross (Figure 1C). After the stimulus period, the two thought probe questions were presented.

Thought Probes

Thought probes are commonly utilized in the study of mind wandering, serving as a means to detect subjective thought states (Hua et al., 2022; Mills et al., 2018; Franklin et al., 2011; Christoff, Gordon, Smallwood, Smith, & Schooler, 2009; Smallwood & Schooler, 2006). Two questions were posed during each probe, addressing two separate dimensions of thought: (1) deliberate constraint and (2) task relatedness. Scores for thought probes were rated on a visual analog scale (VAS) ranging from 0 to 100. Participants were instructed to select a score reflecting their state just before each thought probe. Specifically, for deliberate constraint, participants were asked: "Are you trying to control or pay specific attention to your thought or body movements?", where 0 = *spontaneous* and 100 = *deliberate*. For task relatedness, during the resting state, participants were asked: "Are you keeping your eyes on '+'?", and during the RTT: "Are you focused on the task?", where 0 = *off task* and 100 = *on task* (Figure 1E).

EEG Recording and Preprocessing

EEG data were collected using a 64-channel BrainAmp amplifier (Brain Products, Munich, Germany; International 10–20 system) at a sampling rate of 500 Hz. Fpz was used as the online reference; and Afz, as the ground. The impedance for all electrodes was maintained below 5 kΩ. EEG preprocessing was conducted using the EEGLAB toolbox (Delorme & Makeig, 2004; sccn.ucsd.edu/eeglab/). EEG signals were rereferenced offline to the average of all electrodes. The signals were band-pass zero-phase finite impulse response filtered at 1–50 Hz. Bad channels were excluded using the clean_rawdata EEGLAB extension, with the default settings applied. All stationary artifacts, especially eye movements (blinks and saccades), were removed using independent component analysis and excluded using the MARA EEGLAB extension (Winkler et al., 2014; Winkler, Haufe, & Tangermann, 2011). Data quality was visually verified to ensure that all signals included in subsequent analyses were of good quality and artifact free. Preprocessing procedures (channel cleaning and independent component analysis component removal) did not induce systematic between-participant differences in the number of usable RTT trials. We verified that usable trial counts did not differ across participants and that all reported ITPC effects were unchanged when statistically controlling for per-participant trial count.

EEG Measurements

For EEG analysis, given that the shortest ITI was 5 sec, to avoid edge effects caused by filtering operations, signals were first epoched into 6 sec before thought probes (−6 sec to 1000 msec) for the resting state and from 6 sec prestimulus to 6 sec poststimulus (−6 to 6 sec) for the RTT. The prestimulus period was taken as −4000 to 0 sec.

Peak FS

Peak FS quantifies the changes in the instantaneous phase angle of a particular frequency range (e.g., alpha) of a neural oscillation over time, where higher FS indicates faster changes in oscillatory activity and lower FS reflects slower changes in oscillations (Cohen, 2014b). Both computational modeling and human EEG studies have shown that increases in alpha FS (Hua et al., 2022; Mierau, Klimesch, & Lefebvre, 2017; Cohen, 2014b) are directly related to increased input to a neural network. For example, the strength of alpha FS can track the strength of visual input (Cohen, 2014b). These findings suggest that higher cognitive loads during task-focused thoughts are linked to increased alpha FS, making it a suitable measure for tracking the deliberate control and dynamics of on-task thoughts (Hua et al., 2022; Wolff et al., 2019; Cohen, 2014b). Therefore, in this study, we focused on alpha (8–12 Hz) FS (Rodriguez-Larios & Alaerts, 2019; Wolinski, Cooper, Sauseng, & Romei, 2018; Mierau et al., 2017; Cohen, 2014b). Electrode Fz was selected for FS analysis to be consistent with previous literature (Hua et al., 2022). FS was computed following the method proposed by Cohen (2014b). The preprocessed data were first finite impulse response bandpass filtered with 15% transition zone added to each edge of the filter range, and then a Hilbert transform was done, after which the phase angle time series was extracted. FS is the first derivative of the phase angle time series, and a median filter was applied to reduce nonphysiological noise (Cohen, 2014b). Finally, the mean FS value during the prestimulus period (−4000 to 0 msec) was calculated.

Peak Alpha Frequency

Because FS is derived from the temporal derivative of the phase time series, we conducted a complementary control analysis using an alternative peak frequency estimate based on power rather than phase: peak alpha frequency (PAF; Ramsay, Lynn, Schermitzler, & Sponheim, 2021; Rathee, Bhatia, Punia, & Singh, 2020). To estimate trial-level PAF, we extracted prestimulus EEG data (−4000 to 0 msec relative to stimulus onset) from the Fz electrode. The data were first linearly detrended before spectral analysis. Power spectral density (PSD) was computed for each trial using Welch's method with a 2-sec Hamming window, 50% overlap, and zero-padding to the next power of two. The resulting spectra were averaged across the window segments to obtain a stable PSD estimate for each trial.

For each participant and trial, PAF was defined as the frequency with the maximum PSD within the alpha band (8–12 Hz; Ramsay et al., 2021; Rathee et al., 2020). This procedure yielded a single PAF value per trial, which was stored for subsequent statistical analysis.

SE

SE quantifies the degree of uncertainty and predictability of a time series from one time point to the next (Lau et al., 2022). A high SE value reflects high uncertainty, indicating that the next time point in the series is difficult to predict based on previous time points. Conversely, a low SE value denotes low uncertainty, suggesting that the next time point can be more easily predicted from preceding data. This study focuses on SE because of its established relationship with information input, as suggested in previous literature (Lechner & Northoff, 2023; Lau et al., 2022; Liu et al., 2018). Greater information input leads to increased complexity in neural oscillations, resulting in higher SE values (Lechner & Northoff, 2023; Lau et al., 2022; Liu et al., 2018). Furthermore, SE serves as a forward-looking metric, reflecting the extent to which the next time point in a series can be predicted. This property makes SE particularly relevant for examining how ongoing thoughts influence poststimulus cognition and behavior (Lechner & Northoff, 2023; Liu et al., 2018). SE was computed for the alpha band (8–12 Hz) during the prestimulus period (−4000 to 0 msec).

For a time series $\{x_i\}$, with embedding dimension m, tolerance r, and number of data points N, the sample entropy $SE(m, r, N)$ is defined as the negative natural logarithm of the conditional probability that, given two sequences of length m with a distance less than r, the corresponding sequences of length $m + 1$ will also remain within distance r. The specific calculation procedure of SE is as follows:

1. Construct an m-dimensional vector from the sequence $\{X(i)\}$, that is:

$$X(i) = [x(i), x(i+1), \cdots, x(i+m-1)],$$
$$i = 1 \sim N - m + 1$$

2. Define the distance between $X(i)$ and $X(j)$ as the maximum absolute difference of their corresponding elements:

$$X(i) = [x(i), x(i+1), \cdots, x(i+m-1)],$$
$$i = 1 \sim N - m + 1$$

If the maximum absolute difference of the corresponding elements of $X(i)$ and $X(j)$ is less than r, then for each i, compute the distance between $X(i)$ and all other vectors $X(j), j = 1, 2, \cdots, N - m + 1$.

3. For a given threshold $r > 0$, count the number of times that

$$d[X(i), X(j)] < r$$

holds and calculate the ratio of this number to the total number of vectors $N - m$. Denote it as

$$C_i^m(r) = \frac{1}{N-m}\,\mathrm{num}\{d[X(i), X(j)] < r\},$$
$$i = 1, 2, \cdots, N - m + 1, i \neq j$$

4. Take the average over all i, denoted as $B^m(r)$:

$$B^m(r) = \frac{1}{N-m+1}\sum_{i=1}^{N-m+1} B_i^m(r)$$

5. Increase the embedding dimension by 1 ($m + 1$), and repeat Steps 1–4 to obtain $B_i^{m+1}(r)$ and $B^m(r)$.
6. Theoretically, the SE of the sequence is then defined as

$$SampEn(m, r) = \lim_{N\to\infty}\left\{-\ln\frac{B^{m+1}(r)}{B^m(r)}\right\}$$

However, in practical applications, N is finite. When N takes a limited value, the SE is estimated as

$$SampEn(m, r, N) = -\ln\frac{B^{m+1}(r)}{B^m(r)}$$

Following previous research (Lechner & Northoff, 2023), the embedding dimension (m) was set to 2, indicating that two consecutive data points were utilized for pattern matching. The similarity criterion (r) was established at .2, signifying that data points were deemed a "match" (i.e., indistinguishable) if the absolute amplitude difference between them fell below 20% of the standard deviation of the time series.

ITPC

ITPC quantifies the degree of phase consistency across trials for each participant (Duprez et al., 2021; López et al., 2019; Rapela et al., 2018; van Diepen & Mazaheri, 2018; Cohen, 2014a). ITPC is quantified as the length of the average phase vector across trials:

$$ITPC_{tf} = \left| \frac{1}{n} \sum_{r=1}^{n} e^{ik_{rf}} \right|$$

where n is the number of trials and $e^{ik_{rf}}$ provides the complex polar representation of the phase angle k at time–frequency point tf on trial r. The double bars denote the vector length, reflecting the consistency of phase alignment across trials.

In this study, ITPC was computed for the period from −4000 to 2000 msec relative to stimulus onset during the RTT. The ITPC values from −4000 to −1000 msec served as the baseline, and baseline correction was applied. Subsequently, the average baseline-corrected ITPC value for the poststimulus period (0–1000 msec) was calculated for each participant and used in further analyses. ITPC in the alpha band (8–12 Hz) during the poststimulus period (0–1000 msec) was analyzed at electrode Fz. In addition, ITPC in the theta (4–7 Hz) and delta (1–4 Hz) bands was analyzed as control conditions. Although some researches take theta/beta ratio as a marker of spontaneous thought (van Son, De Blasio, et al., 2019; van Son, de Rover, et al., 2019), some studies take such effect as an artifact caused by the changes of theta and alpha FS (Lansbergen, Arns, van Dongen-Boomsma, Spronk, & Buitelaar, 2011). Therefore, in this study, we mainly focused on the frequency bands below beta and did not take beta into consideration.

For participant-level analyses that used all available RTT trials per participant (target $n = 20$), we computed baseline-corrected ITPC in the alpha band (8–12 Hz) over 0–1000 msec and used these ITPC values directly. For within-participant comparisons after the median split, each bin comprised exactly 10 trials. To equate for small and fixed n, ITPC was converted to ITPCz using the standard transformation with $n = 10$.

Phase Shuffling

To test whether FS and SE during the prestimulus period on poststimulus neural oscillations and associated RT were mediated by phase characteristics, we performed phase shuffling on the EEG time series. This method randomizes the phase of the EEG signals without altering their amplitude (Lechner & Northoff, 2023).

After phase shuffling, the EEG signals were filtered into the alpha band, and we reevaluated prestimulus FS, prestimulus SE, and poststimulus ITPC (see Supplementary Figure 1 for FS and ITPC time series after phase shuffling) across different thought conditions (spontaneous vs. deliberate, off-task vs. on-task). In addition, we calculated Pearson correlations between prestimulus SE and poststimulus ITPC, prestimulus SE and RTs as measured by RT, and poststimulus ITPC and RT. If the observed effects of thought on neural oscillations, and the relationships between prestimulus and poststimulus measures are genuinely phase dependent, all significant effects should disappear after phase is shuffled (i.e., randomized).

Simulation

To examine whether SE and FS are inherently coupled or dissociable, we performed a numerical simulation using synthetic signals with and without scale-free noise.

Pure sinewave condition. We first generated artificial signals consisting of a single pure sinewave. The signal length was fixed at 10 sec, with a sampling rate of 500 Hz, yielding a time vector of 0–10 sec in 1/500-sec steps. Test frequencies ranged from 8 to 20 Hz in 1-Hz increments. For each frequency, a sinusoidal signal was constructed as

$$x(t) = sin(2\pi ft)$$

where f denotes the test frequency. In some cases, a small Gaussian noise (amplitude = 0.01) was added to evaluate robustness. SE was computed for each simulated signal using an embedding dimension $m = 2$ and a tolerance parameter $r = 0.2$ (defined as 20% of the signal's standard deviation). The last SE value was taken as the representative entropy for that frequency.

Sinewave with 1/f noise condition. In a second step, we extended the simulation to more realistic, scale-free signals by adding pink (1/f) noise. Pink noise was generated in the frequency domain (spectral exponent $\alpha = 1$), normalized to unit standard deviation, and scaled to a fixed amplitude of 0.3. Each sinewave (8–20 Hz, 10-sec duration, 500-Hz sampling) was superimposed with the pink noise, yielding

$$x(t) = sin(2\pi ft) + \mathrm{pink}(t)$$

The same SE parameters were applied ($m = 2$, $r = 0.2$), and SE was extracted for each frequency.

Statistical Analysis

Median Split

To perform statistical comparisons between thought types, a median split was applied to each participant's thought probe scores. The top 10 trials with the highest scores were classified as deliberate or on-task trials, whereas the bottom 10 trials were categorized as spontaneous or off-task trials. We then compared RT, prestimulus SE, prestimulus FS, prestimulus PAF, and poststimulus ITPC between on-task and off-task as well as deliberate and spontaneous thoughts using paired t tests. For ITPC comparisons, because each group contained 10 trials after the median split, we transformed ITPC values into ITPCz values using the following formula: $ITPCz = n * ITPC^2$, where n represents the number of trials (Cohen, 2014a).

Comparison between Thought Dimensions

To examine the similarity between the two thought dimensions, task relatedness and deliberate control, we employed three analytical methods. First, we calculated the Pearson correlation between the VAS scores of the two dimensions on both a subject and trial basis (excluding outliers exceeding 3 *SD*s). Second, we applied the Kolmogorov–Smirnov (K-S) test to determine whether the distributions of the two thought dimensions across all trials were significantly different. Third, using the previously described median split, we calculated the number of trials for each participant that fell into the following four categories: (1) on-task and spontaneous, (2) off-task and spontaneous, (3) on-task and deliberate, and (4) off-task and deliberate. Because the counts of these four categories are categorical rather than continuous variables, we employed the nonparametric Friedman test to assess whether there were significant differences among the counts.

If the two dimensions exhibit high similarity, we would expect a strong correlation between their VAS scores and no significant difference between their distributions. In addition, the number of on-task and deliberate trials would be significantly greater than the number of on-task and spontaneous trials, whereas the number of off-task and spontaneous trials would be significantly greater than the number of off-task and deliberate trials.

Mediation Model

We employed a mediation model to examine the relationship between thought, prestimulus FS, and SE. This analysis was conducted at the trial level, meaning all trials across all participants were included in the model. This approach was chosen under the assumption that thoughts vary across trials, and a participant-level analysis, which captures interparticipant variability, would obscure the trial-specific impact of thoughts on the measured variables. In this model, thought VAS scores were included as the independent variable, FS as the mediator, and SE as the dependent variable. In addition, we used a mediation model to investigate how prestimulus SE influences poststimulus RT through poststimulus ITPC as a mediator. For this analysis, a participant-level approach was adopted, as ITPC measures phase coherence across all trials for each participant.

Given that the prestimulus (thought → FS → SE) and poststimulus (SE → ITPC → RT) pathways were analyzed at different hierarchical levels, we additionally applied both single-level and multilevel structural equation modeling approaches to more comprehensively assess the mediation effects along the prestimulus thought–FS–SE pathway.

In the single-level models, mediation was examined exclusively at the trial level (within-participant), treating trials as independent observations while clustering standard errors by participant. These models specified that the experimental condition (deliberate control or task relatedness) predicted the first-stage mediator (FS), which in turn predicted the second-stage mediator (SE). The condition also had a direct effect on SE, allowing for the estimation of both direct and indirect effects. Standard errors were estimated using nonparametric bootstrapping with 2000 resamples to derive confidence intervals for indirect and total effects.

In the multilevel models, we explicitly modeled two levels of data: the trial level (Level 1) and the participant level (Level 2). The Level 1 structure mirrored that of the single-level model, capturing within-participant variability. At Level 2, the same mediation paths were modeled across participants, allowing us to decompose between-participant effects. This hierarchical approach permitted the estimation of cross-level mediation (e.g., whether the effect of deliberate control or task relatedness on FS and SE varied across individuals). These models were estimated using robust Huber–White sandwich estimators to account for nonindependence within participants.

Separate models were estimated for each experimental condition (deliberate control and task relatedness) using the lavaan package in R. All models treated participant as the clustering variable. Final results report standardized path coefficients and 95% confidence intervals for direct, indirect, and total effects at each level.

Resting-state Analysis

To verify that the relationship between thoughts and prestimulus oscillations (FS and SE) was truly task related and not an artifact of thought processes during the resting state, we selected 4000 msec (matching the prestimulus period in the task) before each thought probe in the resting state as pseudo-trials and calculated FS and SE for these pseudo-trials. As in the task condition, we applied a median split on thought VAS scores for each participant to divide trials into spontaneous and deliberate trials (based on deliberate control) and off-task and on-task trials (based on task relatedness). Paired t tests were then conducted to compare FS and SE between spontaneous and deliberate trials as well as between off-task and on-task trials in these pseudo-trials.

RESULTS

Both Thought Dimensions (Deliberate Control and Task Relatedness) Shape Task-related Behavior (RT) and Are Closely Related to Each Other

The nature of thought impacts behavioral performance during cognitive tasks (e.g., RT) (Bastian & Sackur, 2013; Franklin et al., 2011; Smallwood, Beach, Schooler, &

Handy, 2008; Smallwood & Schooler, 2006). Here, we tested whether the thought dimensions, deliberate control, and task relatedness are important determinants of performance. We first employed paired t tests to examine whether RT during the RTT differed between spontaneous and deliberate thoughts as well as between off-task and on-task thoughts. RT during deliberate thoughts was significantly shorter than that during spontaneous thoughts, $t(39) = 3.47$, $p = .001$ (Figure 3A). Likewise, RT during on-task thoughts was shorter as compared to that during off-task thoughts, $t(39) = 3.69$, $p = .001$ (Figure 3B). This pattern of results raises the question of whether the thought dimensions of task relatedness and deliberate control are linked to one another. To address this, we first applied Pearson correlation analysis to the VAS scores of these two dimensions at both the participant level (averaging VAS values for each participant before correlation) and trial based (correlating all trials across all participants). Results revealed that task relatedness (on-task vs. off-task) was highly correlated with deliberate control (spontaneous vs. deliberate) at both levels (participant level: $r(38) = .80$, $p < .001$ [Figure 3C]; trial based: $r(798) = .71$, $p < .001$ [Figure 3D]).

Furthermore, we conducted a two-sample K-S test to compare the distributions of both task relatedness and deliberate control across all trials. The results showed no significant difference between the distributions of these two thought dimensions ($D = 0.06$, $p = .178$; Figure 3E), indicating no significant difference in terms of their distributions. This further confirms the close relationship between the two thought dimensions.

Finally, we calculated the number of trials for each participant that fell into the following four combinations: (1) spontaneous and on-task, (2) spontaneous and off-task, (3) deliberate and on-task, and (4) deliberate and off-task. We then tested whether the counts differed significantly using a nonparametric Friedman test. Results indicated significant differences among the four groups ($p < .001$; Figure 3F). Post hoc comparisons with Dunn's multiple comparisons test revealed that the trial numbers of spontaneous/off-task and spontaneous/on-task thoughts differed significantly as did spontaneous/off-task and

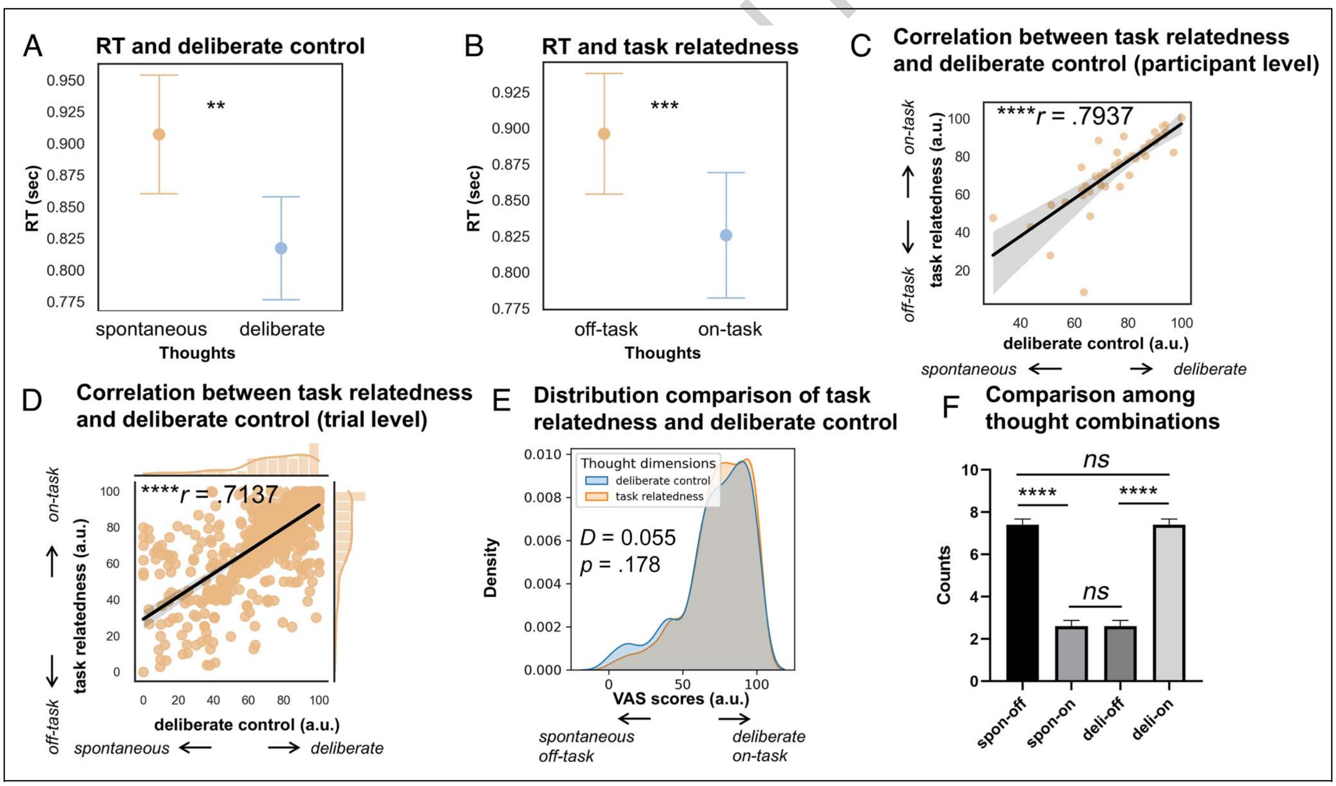

Figure 3. Behavioral analysis of RTs and thought dimensions. (A) RTs and deliberate control. RTs were significantly shorter during deliberate thoughts compared to spontaneous thoughts. (B) RTs and task relatedness. RTs were significantly shorter during on-task thoughts compared to off-task thoughts. (C) Correlation between task relatedness and deliberate control (participant level). Task relatedness and deliberate control showed a strong positive correlation at the participant level. (D) Correlation between task relatedness and deliberate control (trial level). Task relatedness and deliberate control were also strongly correlated at the trial level. (E) Distribution comparison of task relatedness and deliberate control. No significant differences were found between the distributions of these two dimensions across trials. (F) Comparison among thought combinations. The distribution of trials among the four thought combinations revealed significant differences, highlighting the predominant co-occurrence of on-task and deliberate thoughts as well as of off-task thoughts with spontaneous thoughts. *$p < .05$, **$p < .01$, ***$p < .001$, ****$p < 0.0001$. *ns* = not significant; spon = spontaneous; deli = deliberate.

deliberate/off-task thoughts (all $ps < .001$; Figure 3F). These results indicate an association of spontaneous thoughts with off-task thoughts than with on-task thoughts, whereas deliberate thoughts are more linked to on-task thoughts than off-task thoughts. In contrast, spontaneous/off-task and deliberate/on-task thoughts were statistically indistinguishable (all $ps > .999$; Figure 3F); this suggests a close relationship between the two thought dimensions at their extreme ends.

In conclusion, these behavioral findings suggest that the thought dimensions of deliberate control and task relatedness are both associated with task-related behavior, namely, RT. More deliberate and on-task thoughts are linked to shorter RTs, whereas spontaneous and off-task thoughts are associated with longer RTs, thus highlighting their interrelatedness. This close association was further substantiated by the observation that spontaneous thoughts were more strongly associated with off-task thoughts, whereas deliberate thoughts were more closely linked to on-task thoughts. Therefore, at the behavioral level, the two thought dimensions are closely coupled with each other.

On-task and Deliberate Thoughts Show Faster and Less Predictable Neural Dynamics during Prestimulus Activity

In the previous section, we demonstrated the close relationship of the two thought dimensions (task relatedness and deliberate control) at the behavioral level. Does this also hold at the neuronal level? In that case, one would expect that these thought dimensions are also not dissociable in terms of relevant neural measures. Paired t tests (Figure 4B) showed that prestimulus FS for spontaneous versus deliberate control, $t(39) = 2.28$, $p = .028$, and on-task versus off-task, $t(39) = 2.29$, $p = .027$, significantly differed, as did SE for spontaneous versus deliberate control, $t(39) = 2.83$, $p = .007$, and on-task versus off-task, $t(39) = 3.191$, $p = .0028$.

Given their similar direction with respect to the two thought dimensions, we next correlated prestimulus FS with SE. As expected, Pearson correlation analysis revealed that FS and SE were significantly correlated during the prestimulus period, $r(37) = .49$, $p = .002$ (Figure 4C). Given previous literature showing that changes of SE

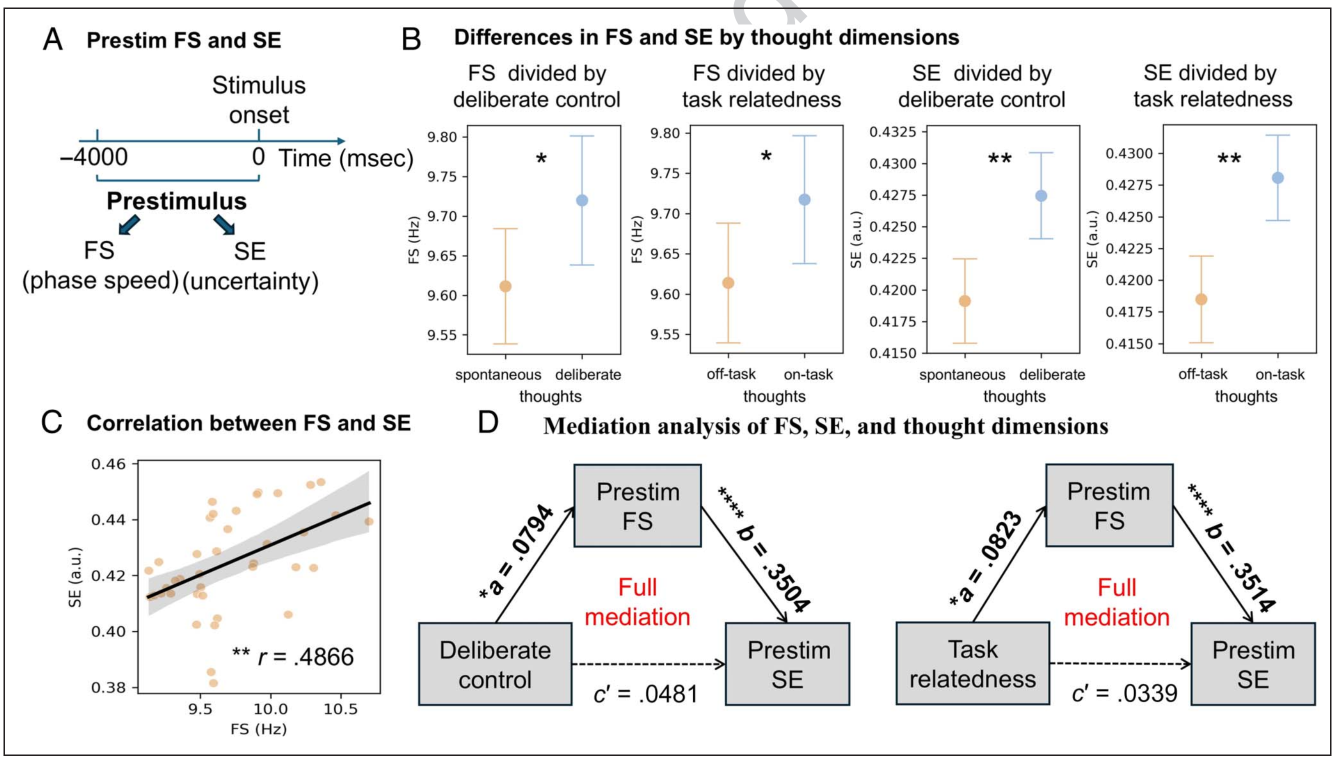

Figure 4. Prestimulus neuro-oscillations dynamics and their relationship with thought dimensions. (A) Prestimulus FS and SE. The average alpha-band (8–12 Hz) FS and SE over the prestimulus period (−4000 to 0 msec) was calculated to examine neural oscillations preceding stimulus onset. (B) Differences in FS and SE by thought dimensions. Both prestimulus FS and SE were significantly higher during on-task and deliberate thoughts compared to off-task and spontaneous thoughts. (C) Correlation between FS and SE. A significant positive correlation was observed between prestimulus FS and SE in the alpha band, highlighting their interplay during the prestimulus period. (D) Mediation analysis of FS, SE, and thought dimensions. Mediation models revealed that prestimulus FS fully mediated the relationship between thought dimensions (deliberate control and task relatedness) and prestimulus SE, indicating a tight linkage between ongoing neuro-oscillation dynamics and ongoing thought. *$p < .05$, **$p < .01$, ****$p < .0001$. In the mediation models, solid lines indicate a significant effect, whereas dashed lines represent a nonsignificant effect. Prestim = prestimulus.

could be taken as consequence of FS (Aboy, Cuesta-Frau, Austin, & Mico-Tormos, 2007), we applied a mediation model to analyze the relationships among thought VAS scores, prestimulus FS, and prestimulus SE. The results indicated full mediation effects for both thought dimensions (deliberate control: $a = 0.079$ [$p = .025$], $b = 0.350$ [$p < .001$], $c' = 0.048$ [$p = .148$], $c = 0.076$ [$p = .032$]; task relatedness: $a = 0.082$ [$p = .020$], $b = 0.351$ [$p < .001$], $c' = 0.034$ [$p = .308$], $c = 0.063$ [$p = .076$]; Figure 4D). Moreover, no mediation effect can be found when thought dimensions were taken as mediators (deliberate control: $a = 0.079$ [$p = .025$], $b = 0.048$ [$p = .148$], $c' = 0.350$ [$p < .001$], $c = 0.354$ [$p < .001$]; task relatedness: $a = 0.082$ [$p = .020$], $b = 0.034$ [$p = .308$], $c' = 0.351$ [$p < .001$], $c = 0.354$ [$p < .001$]; Supplementary Figure 2) or dependent variables (deliberate control: $a = 0.354$ [$p < .001$], $b = 0.055$ [$p = .148$], $c' = 0.060$ [$p = .112$], $c = 0.079$ [$p = .025$]; task relatedness: $a = 0.354$ [$p < .001$], $b = 0.039$ [$p = .308$], $c' = 0.069$ [$p = .069$]; Supplementary Figure 2). Together, these findings suggest that the impact of both thought dimensions on prestimulus entropy is fully mediated by FS. Specifically, the neural impact of the two thought dimensions on uncertainty (SE) is mediated by the phase-based measurement of neural speed (FS).

To further validate this mechanistic relationship, we conducted a supplementary simulation using synthetic signals with and without scale-free noise (Supplementary Figure 3). When the signal consisted of a pure sinewave, increasing frequency did not lead to any systematic change in SE, indicating that SE and FS were independent for purely rhythmic dynamics (Supplementary Figure 3). However, when $1/f$ noise was added—approximating the broadband characteristics of real EEG—SE exhibited a clear upward trend with increasing frequency (Supplementary Figure 3). This pattern demonstrates that the coupling between FS and SE emerges only in the presence of scale-free background dynamics. Hence, the observed FS–SE relationship in EEG likely reflects the contribution of scale-free neural processes, which provide the background that enables oscillatory speed to modulate entropy (Ao, Klar, Catal, Wang, & Northoff, 2025).

Given that FS itself is phase based (Cohen, 2014), we tested for the impact of the phase-related processes on our prestimulus measures' relation with the two thought dimensions. To test whether the relationships between prestimulus FS and SE with the two thought dimensions are phase dependent (rather than just amplitude based), we conducted paired t tests on phase-shuffled prestimulus FS and SE. Unlike when phase is unshuffled, the results showed no significant differences between off-task and on-task thoughts or between spontaneous and deliberate thoughts after phase shuffling (all ps $> .16$; Supplementary Figure 4). As an additional control analysis, we also employed an alternative method for estimating peak frequency based on power rather than phase, namely, PAF, and tested whether it differed across thought conditions. Paired t tests revealed that this power-based measure showed no significant differences between thought states (deliberate control: $t(39) = 0.33$, $p_{\mathrm{FDR}} = 1.000$; task relatedness: $t(39) = 1.69$, $p_{\mathrm{FDR}} = 1.000$). These findings suggest that the impact of both prestimulus FS and SE on the two thought dimensions is mediated by their underlying phase-related processes.

Are the observed relationships of prestimulus dynamics with the two thought dimensions associated with the task context in which they are set? To test for that, we conducted the same analyses, for example, paired t tests on FS and SE, in resting-state pseudo-trials, which has the same timing as in the prestimulus of task. To verify whether the relationship between thought and prestimulus FS and SE was task related, we conducted paired t tests on FS and SE in resting-state pseudo-trials. The results showed no significant differences in either FS or SE between spontaneous and deliberate thoughts or between off-task and on-task thoughts in resting-state pseudo-trials (FS deliberate vs. spontaneous thought: $t(39) = 0.50$, $p = .618$; FS on-task vs. off-task: $t(39) = 0.14$, $p = .888$; SE deliberate vs. spontaneous: $t(39) = 0.63$, $p = .530$; SE on-task vs. off-task: $t(39) = 1.95$, $p = .058$; Supplementary Figure 5). This suggests that the relationship of the impact of the prestimulus dynamics on the two thought dimensions is associated with the task context as it does not occur during the absence of a task during the resting state.

Together, these findings suggest that both on-task and deliberate thoughts accelerate phase-based neural speed (FS) and, at the same time, increase the uncertainty/flexibility (SE) of the neural oscillations during the prestimulus period. The supplementary simulation further demonstrates that this FS–SE coupling originates from scale-free neural dynamics, providing a mechanistic account for how different types of thought (on-task/deliberate vs. off-task/spontaneous) exert distinct influences on prestimulus neural activity.

To further examine the mechanistic role of FS in linking thought dimensions with prestimulus SE, we conducted mediation analyses separately for deliberate control and task relatedness. In the single-level models with cluster-robust standard errors, both dimensions showed significant indirect effects on SE through FS. Deliberate control positively predicted FS ($a = 0.002$, $p = .013$), and FS in turn predicted SE ($b = 0.022$, $p < .001$), whereas the direct effect of deliberate control on SE was not significant ($c' = 0.000$, $p = .160$). The indirect effect was significant ($p = .016$), yielding a significant total effect ($p = .036$). A highly similar pattern was observed for task relatedness: Probe 2 positively predicted FS ($a = 0.003$, $p = .013$), and FS predicted SE ($b = 0.022$, $p < .001$), whereas the direct path to SE was again not significant ($c' = 0.000$, $p = .273$). The indirect effect reached significance ($p = .016$), and the total effect was marginally significant ($p = .055$). These results indicate that, at the trial level, both deliberate control and task relatedness influence SE indirectly through FS, with FS serving as a full mediator.

In contrast, multilevel mediation analyses that separated within-participant and between-participant variance did not yield significant indirect effects for either dimension (all ps > .2). At both levels, the paths from deliberate control and task relatedness to FS and SE were nonsignificant, whereas the paths from FS to SE remained robust (for both dimensions; within: $b = 0.020$, $p < .001$; between: $B = 0.025$, $p < .001$). This pattern suggests that the indirect effects detected in the single-level models largely reflect trial-level variability, whereas FS → SE constitutes the most consistent and reliable pathway across analytic levels.

Prestimulus SE Affects RT Mediated by Poststimulus ITPC

We show that both thought dimensions modulate task-related RT while, at the same time, being related to prestimulus activity neurodynamics, for example, FS and SE. This raises the question whether and how the prestimulus dynamics affects poststimulus RT. Given the randomized intervals between stimuli in our task paradigm, we propose that higher ITPC reflects a more consistent internal brain state, wherein participants are in a sustained on-task and deliberate mode, allowing neural oscillations to align optimally just before stimulus onset. Therefore, we calculated phase-based ITPC in alpha band (8–12 Hz) for the period of 0–1000 msec (see Figure 5A and B). First, we applied Pearson correlation analysis between poststimulus ITPC and RT. The results showed a significant correlation, $r(36) = -.38, p = .018$ (Figure 5C): The higher ITPC in the alpha band, the shorter RT. As a control, we also calculated Pearson correlations between poststimulus ITPC in the delta (1–4 Hz) and theta (4–7 Hz) bands with RT, neither of which showed significant correlations (delta: $r(37) = -.18, p = .293$; theta: $r(37) = -.28, p = .085$; Supplementary Figure 6). Furthermore, although ITPC was related to RT, it did not differ across either thought dimension when using median splits and ITPCz (deliberate control: $t(39) = 1.18$, $p = .245$; task relatedness: $t(39) = 1.53$, $p = .134$; Supplementary Figure 7). This outcome raises an important question regarding how prestimulus neurodynamics influence the subsequent poststimulus period in both its neural and behavioral measures.

To test whether prestimulus neurodynamics affect poststimulus RT, we calculated the Pearson correlation

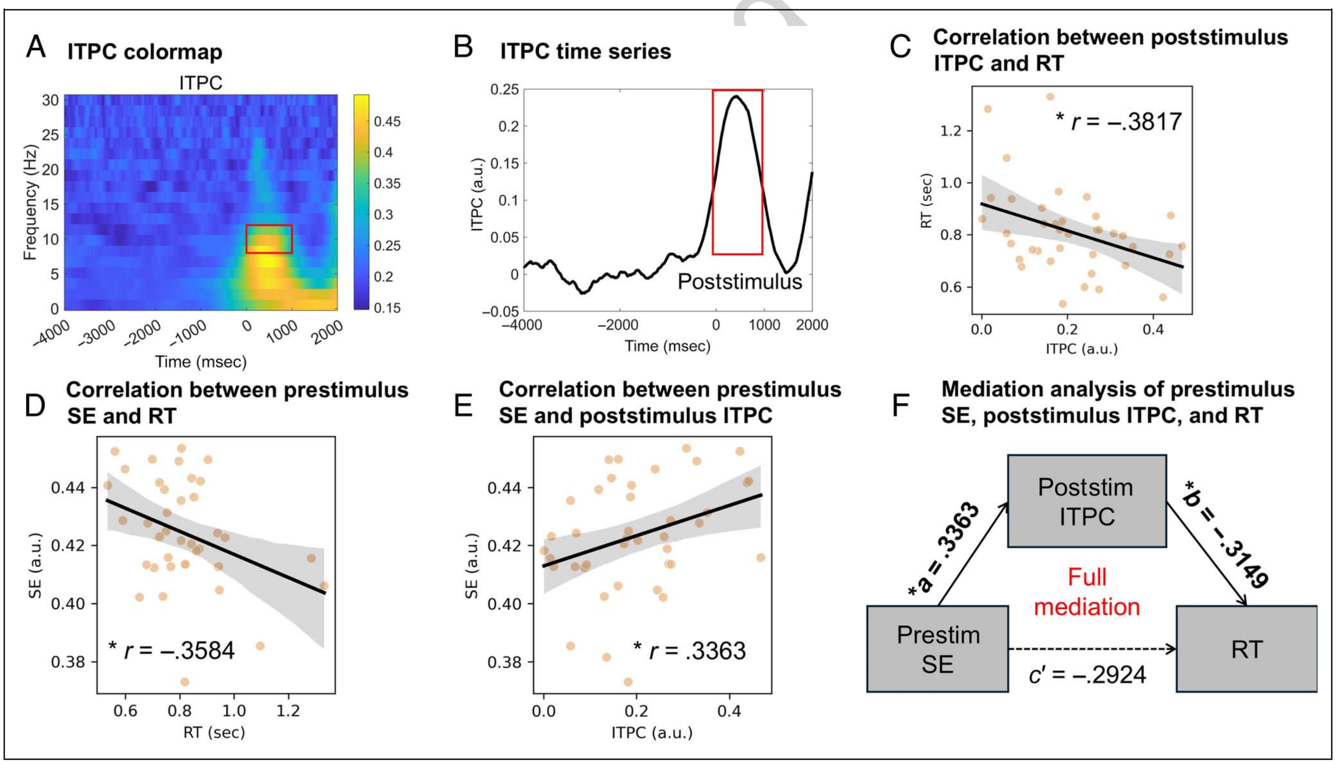

Figure 5. Relationships between ongoing thought and neural activity, poststimulus neural activity, and RT. (A, B) Poststimulus ITPC in the alpha band. Poststimulus ITPC in the alpha band (8–12 Hz) was calculated for the 0- to 1000-msec period to assess neural coherence after stimulus onset. (C) Correlation between poststimulus ITPC and RT. A significant negative correlation was observed, indicating that higher poststimulus ITPC is associated with faster RTs. (D) Correlation between prestimulus SE and RT. Prestimulus SE in the alpha band was significantly negatively correlated with RT, suggesting that prestimulus neural oscillations influence response speed. (E) Correlation between prestimulus SE and poststimulus ITPC. Prestimulus SE was significantly positively correlated with poststimulus ITPC, linking prestimulus dynamics to poststimulus neural activity. (F) Mediation analysis of prestimulus SE, poststimulus ITPC, and RT. A mediation model revealed that poststimulus ITPC significantly mediated the relationship between prestimulus SE and RT, demonstrating that prestimulus neuro-oscillation dynamics influence response speed via poststimulus neural synchronization (ITPC). *p < .05. *ns* = not significant; Poststim = poststimulus; Prestim = prestimulus.

between prestimulus SE and RT. The results show a significant correlation, $r(37) = -.36$, $p = .027$ (Figure 5D), whereby higher prestimulus SE was associated with shorter RT in the poststimulus period. Moreover, prestimulus SE was also significantly correlated with poststimulus ITPC, $r(39) = .34$, $p = .034$ (Figure 5E), whereby higher prestimulus entropy was associated with higher poststimulus ITPC.

As a control, we also calculated Pearson correlations on phase-shuffled data between poststimulus alpha ITPC and RT, prestimulus SE and poststimulus ITPC, and prestimulus SE and RT. The results showed no longer any significant correlations (all ps $> .3$; Supplementary Figure 8). This suggests the key relevance of phase-based processes for connecting prestimulus dynamics with poststimulus coherence and behavior.

To further specify the prestimulus–poststimulus relationships, we applied a mediation model among prestimulus SE, poststimulus ITPC, and RT. The results showed a significant full mediation effect ($a = 0.336$, $p = .034$; $b = -0.315$, $p = .045$; $c' = -0.292$, $p = .061$; $c = -0.398$, $p = .011$; Figure 5F). The impact of prestimulus neural dynamics (SE), on poststimulus behavior (RT), is fully mediated by the degree of phase coherence (ITPC) at stimulus onset. In contrast to SE, no such mediation model was observed for prestimulus FS ($a = 0.185$, $p = .254$; $b = -0.431$, $p = .007$; $c' = 0.098$, $p = .523$; $c = 0.018$, $p = .912$; Supplementary Figure 9). This suggests a key role for SE (but not for FS) in directly connecting prestimulus and poststimulus activity.

Finally, we applied mediation models among prestimulus SE and FS with poststimulus ITPC and RT respectively on phase shuffled data; results show no mediation effect was found (SE: $a = -0.013$ [$p = .935$], $b = -0.006$ [$p = .973$], $c' = 0.057$ [$p = .731$], $c = 0.057$ [$p = .727$]; FS: $a = 0.077$ [$p = .637$], $b = -0.006$ [$p = .971$], $c' = -0.004$ [$p = .983$], $c = -0.004$ [$p = .981$]; Supplementary Figure 10).

Together, the findings show a key role for phase-based processes in mediating the impact of prestimulus dynamics on poststimulus activity and behavior. Specifically, prestimulus entropy modulates poststimulus RT through the phase-based ITPC. These results suggest that the impact of task relatedness and executive control on task-related neural activity and its behavior is mediated by their modulation of prestimulus phase-based neural dynamics (FS, SE). In addition, through phase-based ITPC, this is carried over to the subsequent poststimulus period.

Decoupling of Thought Dimensions during the Resting State Abolishes Their Differential Modulation of Neurodynamics

We demonstrate clear differences in the prestimulus neuro-oscillation dynamics between on-task deliberate and off-task spontaneous thoughts. This raises the question of where and how such differences originate. This raises the question whether prediction/anticipation plays a key role in differentiating on-task deliberate and off-task spontaneous thoughts in their prestimulus neuro-oscillation dynamics. Although we did not include direct measures of prediction/anticipation in our design, we can nevertheless address this question in an indirect way, that is, through comparing task state with the resting state.

Paired t tests were conducted to examine differences in two thought dimensions (measured by VAS) between the resting-state and task conditions. The results showed that deliberate control scores were significantly lower in the resting state than during the task condition, $t(39) = 2.98$, $p_{\mathrm{FDR}} = .010$ (Figure 6A). In contrast, there were no significant differences in task-relatedness scores between resting and task states, $t(39) = 0.90$, $p_{\mathrm{FDR}} = 1.000$ (Figure 6B).

Given the high correlation between the two dimensions (deliberate control and task relatedness) under the task condition, we also investigated their relationship during the resting state. Pearson correlations were computed separately for trials and individuals in the resting state, and the distributions of trials across the two dimensions were compared by applying K-S test. Despite a high correlation also observed during the resting state (participant based: $r(38) = .73$, $p < .001$; trial based: $r(798) = .57$, $p < .001$; Supplementary Figure 11), the distribution of the two thought dimensions was significantly distinct between each other during the resting state (but not during the task state; see above; $D = 0.13$, $p < .001$ (Figure 6C). We then compared the two thought dimensions' correlation coefficients between resting and task states: Pearson and Filon's z test showed a significant difference in their correlation coefficients between task and resting state conditions, with the latter being significantly lower than the former ($z = -5.11$, $p < .001$). To quantify the two thought dimensions' degree of decoupling during the resting state (compared to the task state), we calculated the absolute difference in their correlation coefficients as a measure of coupling strength; then, we performed a permutation test (1000 iterations) to assess whether the observed difference was significantly greater than random noise. Results demonstrated that the task-resting state difference in their coupling strength between thought dimensions (as measured by the absolute value of correlation coefficients) was indeed significant (real difference = 0.113; permutation mean difference = −0.001, 95% CI [−0.0627, 0.0631], $p = 0.001$; Figure 6D). These findings suggest a certain degree of decoupling between the two thought dimensions during the resting state compared to the task state.

Finally, we investigated whether this decoupling affected the relationship between thought dimensions and prestimulus neuronal dynamics (FS and SE). During the resting state, no significant differences were observed in FS or SE across either dimension—deliberate control (FS: $t(39) = 0.50$, $p = .618$; SE: $t(39) = 0.63$, $p = .530$; Figure 6E and F) or task relatedness (FS: $t(39) = 0.14$, $p = .888$; SE: $t(39) = 1.95$, $p = .058$; Figure 6G and H).

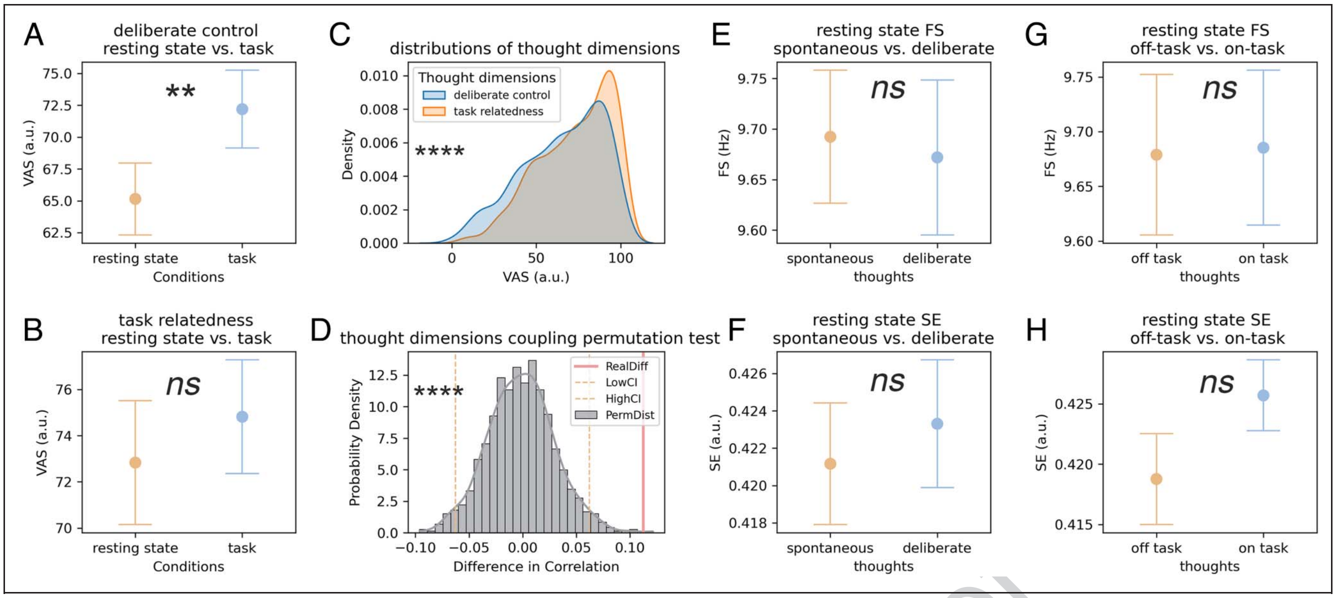

Figure 6. Reduced coupling between thought dimensions (deliberate control and task relatedness) in the resting state and relationship to neuro-oscillation dynamics (FS and SE). (A) Comparison of deliberate control between task and resting state. Deliberate control ratings increase significantly during task, indicating greater intentional regulation of thoughts (more deliberate) when anticipating stimulus onset. (B) Comparison of task relatedness between task and resting state. Task relatedness remains similar across resting-state and task conditions, suggesting that the relevance of thoughts to an external goal does not change substantially. (C) Distributions of the two thought dimensions are significantly different from each other during task and resting states. Distributions of the two thought dimensions, deliberate control and task relatedness, differ significantly from each other in resting state, implying weaker coupling when participants are not preparing for a task. (D) Reduced coupling between the two thought dimensions in resting state compared to task condition. Comparing the two thought dimensions' correlation coefficients reveals a significant decrease in their correlation strength during the resting state compared to the task state, confirmed by a permutation test. (E–H) No neuro-oscillation dynamical differences (FS and SE) among the thought dimensions at rest. Neither FS nor SE shows significant changes across varying levels of deliberate control (E–F) and task relatedness (G–H) in the resting state, suggesting heightened FS and SE only emerge when deliberate control and task relatedness are tightly coupled, as in the task condition.

Although tentative, these results indicate that the heightened FS (acceleration in phase-based speed) and SE (increase in uncertainty) occur only during the task condition with its tight coupling of the two thought dimensions (deliberate control and task relatedness).

DISCUSSION

This study explored how two thought dimensions, namely, deliberate control and task relatedness, shape prestimulus and poststimulus neuronal phase dynamics and task-related behavior. We found that, during the prestimulus period, on-task and deliberate thoughts were associated with higher FS and SE, compared to off-task and spontaneous thoughts. This suggests that distinct types of thought are characterized by unique neuronal dynamic signatures in their phase and periodicity of the EEG signal. A mediation model revealed that FS fully mediated the effect of thought dimensions on SE, suggesting that neuro-oscillation speed (FS) drives neural uncertainty/flexibility (SE). Thus, the phase of the signal appears to be the important factor in understanding the relationship to the task relatedness and spontaneity of thought processes.

Furthermore, we observed increased SE during the prestimulus period, which was associated with enhanced phase alignment at stimulus onset. This ultimately facilitated faster RT.

Although the resting state data emphasized that deliberate control and task relatedness may constitute dissociable dimensions of thought, our task data showed that they were strongly coupled in a different context, for example, task rather than rest. This suggests that, under task demands, these two dimensions may converge onto a common underlying mechanism in externally constrained contexts (e.g., executive control engaged to sustain task focus), whereas their relationship may be rather loose in unconstrained contexts such as the resting state. This task–resting difference indicates that the dissociation of thought dimensions may be state and context dependent rather than universal, underscoring the importance of investigating the shared mechanisms that jointly shape both dimensions during task state.

Together, our findings support the possibility of a sequential influence of thought on behavior via phase-based neural dynamics but do not establish causality. Importantly, the phase shuffling analysis revealed that the phase state is the critical driver of these relationships, underscoring the central role of phase dynamics in linking thought over prestimulus dynamics to poststimulus activity and the associated behavior. The following sections provide further discussion about the meaning of these findings in terms of cognition and behavior as well as their underlying neuronal phase-related mechanisms.

From Thoughts to Prestimulus Neural Dynamics: Mechanistic Insights

FS represents the speed of neural oscillation, reflecting the rate of change of phase over time (Hua et al., 2022; Cohen, 2014b). Particularly for the alpha band, faster FS enables the brain to more efficiently process input information (Hua et al., 2022; Mierau et al., 2017; Cohen, 2014b). On-task and deliberate thoughts are associated with higher FS, indexing faster neural speed, facilitating enhanced cognitive control and preparation for the incoming stimulus. This aligns with prior findings that the contents of both perception and thought are linked to phase-based oscillatory cycles in alpha frequencies, where faster neural speeds facilitate more efficient neural synchronization and more deliberate and on-task thoughts (Hua et al., 2022; Wolff et al., 2019; Mierau et al., 2017; Cohen, 2014b).

SE quantifies the unpredictability of neural oscillations, capturing the brain's dynamic readiness for external demands (Lechner & Northoff, 2023; Lau et al., 2022). Higher SE is thus crucial for facilitating flexibility to respond to external stimuli (Lau et al., 2022; Saxe, Calderone, & Morales, 2018; Pedersen, Omidvarnia, Walz, Zalesky, & Jackson, 2017). On-task and deliberate thoughts exhibit higher SE, suggesting enhanced flexibility in the preparatory state from a neuronal perspective (Lau et al., 2022; Saxe et al., 2018; Pedersen et al., 2017). Our mediation model underscores the relationship between FS and SE, where FS serves as a precursor driving neural flexibility. Deliberate and on-task thoughts increase FS, which subsequently raises SE, creating a dynamic prestimulus neural state that supports subsequent phase alignment during task execution. Notably, both thought dimensions, that is, deliberate control and task relatedness, converged on the same mechanistic pathway: Their influence on SE was fully mediated by FS. This trial-level mediation effect was statistically significant for both dimensions in the single-level models, highlighting the central role of phase dynamics, for example, FS as a commonly shared neural mechanism through which different dimensions of thought shape prestimulus neural speed and flexibility. Importantly, these indirect effects were no longer significant in multilevel models that explicitly separated within-participant and between-participant variance, although the FS → SE pathway remained robust across levels. This discrepancy indicates that the apparent mediation likely reflects trial-to-trial variability; this is meaningful given that thoughts fluctuate on a trial-by-trial basis, although this effect may be obscured when aggregating and averaging across participants. Thus, although multilevel analyses provide a more conservative test, the significant trial-based mediation underscores the functional role of FS in linking both deliberate control and task relatedness to SE in a moment-to-moment way, that is, trial based.

From Prestimulus Neuronal Dynamics to Poststimulus Neural Activity and Behavior

The optimal phase alignment at stimulus onset is essential for effective poststimulus synchronization (van Diepen & Mazaheri, 2018). High SE enhances flexibility of neural oscillations, leading to higher phase alignment to the optimal phase and thus higher ITPC and, consequently, shorter RT. In contrast, lower ITPC, stemming from disrupted phase dynamics, slows RTs. The cascading effects of prestimulus phase-based amplitude variability (SE) to ITPC and RT further highlight the pivotal role of phase dynamics in linking thought through neural readiness, and, ultimately, to behavior.

Together, these findings support the plausibility of a mechanistic pathway from thought to behavior: ongoing thought → prestimulus FS → prestimulus SE → poststimulus ITPC → poststimulus RT. This pathway illustrates how ongoing thoughts influence prestimulus phase dynamics, which in turn modulate poststimulus phase synchronization and behavioral performance. Moreover, as supported by our control analyses like phase shuffling, this study provides compelling evidence that phase dynamics, rather than the magnitude of the signal, are critical in linking characteristics of thought to task performance. FS determines the changes in speed of phase cycles, whereas SE captures phase-based amplitude variability, both of which are crucial for achieving optimal phase alignment at stimulus onset. The critical role of phase was further validated by phase-shuffling analyses, which disrupted key relationships and confirmed that phase continuity is essential for neural synchronization and behavioral efficiency (Figure 7).

Limitations

Because of time constraints of 10 min, to ensure a longer prestimulus period to generate and assess ongoing thought, each participant completed only 20 trials in the RTT, similar to previous studies (Rostami et al., 2022; Vanhaudenhuyse et al., 2011). Although ITPC analysis can be conducted based on this number of trials (Cohen, 2014a), it is generally believed that a higher number of trials would better ensure the robustness of ITPC. Although the number of trials was sufficient for the current investigation, in future studies, increasing the number of trials could help improve the reliability of ITPC.

In addition, we employed the relatively simple RTT paradigm. Future studies should adopt more sophisticated paradigms, such as the Stroop task, to enable a more fine-grained investigation of higher-level cognitive processes.

An additional limitation concerns the mediation results. In single-trial analyses, both deliberate control and task

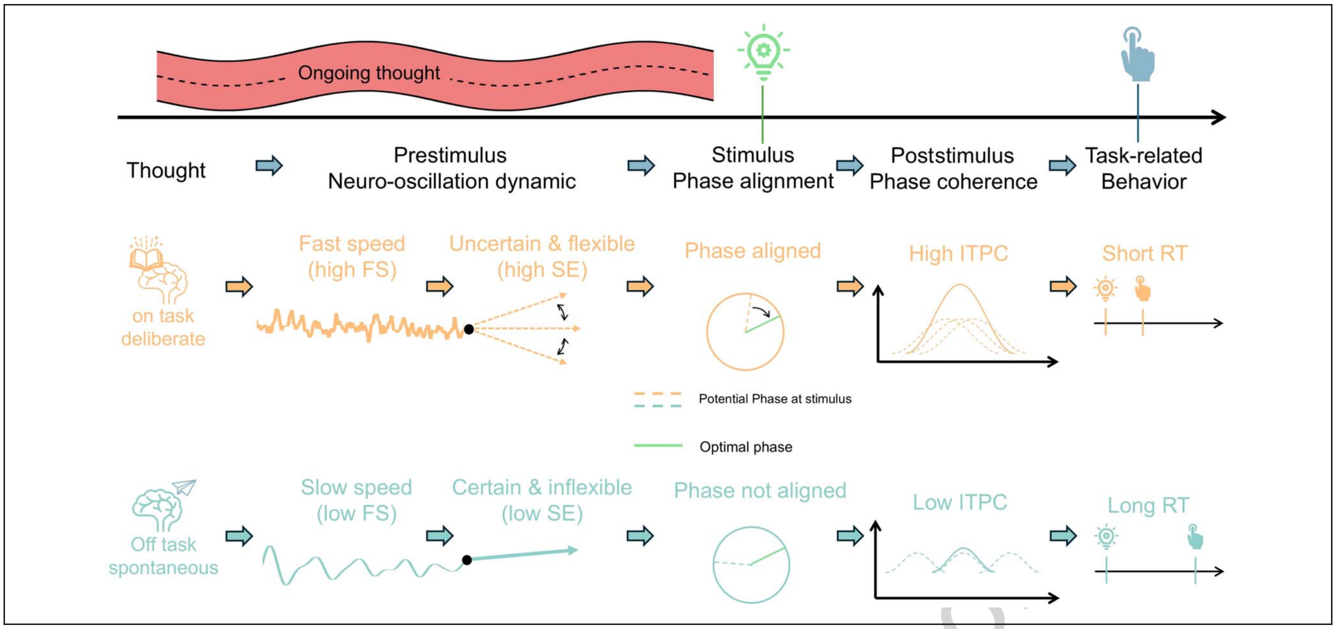

Figure 7. Mechanistic pathway linking thought dimensions to behavioral outcomes through neural dynamics. This figure illustrates the relationship between thought dimensions, prestimulus neural oscillations, poststimulus phase coherence, and RT. Thought dimensions: On-task deliberate thoughts (orange) are associated with faster RTs, whereas off-task spontaneous thoughts (blue) correspond to slower RTs, highlighting their influence on task performance. Prestimulus neuro-oscillation dynamics: On-task deliberate thoughts enhance prestimulus FS (neural phase-based speed) and SE (phase uncertainty). Higher FS enables faster neural oscillations, whereas increased SE facilitates more flexible neural oscillations, preparing for optimal phase alignment at stimulus onset. Conversely, off-task spontaneous thoughts result in lower prestimulus FS and SE, characterized by slower oscillations and reduced flexibility, diminishing the potential for phase alignment. Phase alignment and post-stimulus dynamics: High FS and SE in the prestimulus phase increase the probability of higher phase alignment at stimulus onset, leading to higher poststimulus ITPC in the poststimulus phase. This phase synchronization enhances task performance by reducing RT. Low FS and SE result in lower phase alignment at stimulus onset, leading to lower ITPC and slower RTs. Behavioral outcome: The pathway "ongoing thought → prestimulus FS → prestimulus SE → poststimulus ITPC → poststimulus RT" emphasizes how thought states (on-task deliberate vs. off-task spontaneous) modulate prestimulus neural preparation and poststimulus task execution, ultimately affecting behavioral responses.

relatedness significantly influenced SE through FS, whereas in multilevel models, these indirect effects were not significant at either the within- or between-participant level. This divergence suggests that the mediation pathway is most pronounced when capturing trial-level variability but may be attenuated once one averages across trials as when the hierarchical structure is modeled. However, the number of clusters/participants was relatively small (n = 40), for which reason our between-participant power was limited; this could also contribute to the absence of significance of the mediation analyses, especially those conducted at the participant level, which showed relatively modest post hoc power estimates (approximately 0.45–0.56). Although these values are below the conventional threshold for strong power, they still fall within a range considered adequate for exploratory or theory-driven models. Future work with larger samples and more trials will be necessary to determine whether these trial-level mediation effects generalize robustly across analytic levels.

Conclusion

This study suggests a plausible pathway linking distinct dimensions of thought to behavior through prestimulus and poststimulus neural phase dynamics, although further work is needed to confirm the full mechanistic sequence. On-task deliberate thoughts enhance phase-based FS and SE during the prestimulus period. This, in turn, enhances the flexibility of neural oscillation and thus increases the likelihood of higher coherence of the ongoing phase dynamics with the external stimulus, as manifested in higher poststimulus ITPC. This cascade results in faster RT, emphasizing the key role of the brain's neuronal dynamics in connecting ongoing thoughts to behavior through prestimulus–poststimulus phase dynamics. Our findings demonstrate novel insights into how our ongoing thoughts modulate task-related behavior and cognition.

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Data Availability Statement

The data and analysis scripts that support the findings of this study are not publicly available because of participant privacy concerns. However, they are available from the corresponding author upon reasonable request via e-mail. Supplemental material can be accessed on this article's homepage: https://doi.org/10.1162/JOCN.a.2417.

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Diversity in Citation Practices

Retrospective analysis of the citations in every article published in this journal from 2010 to 2021 reveals a persistent pattern of gender imbalance: Although the proportions of authorship teams (categorized by estimated gender identification of first author/last author) publishing in the *Journal of Cognitive Neuroscience* (*JoCN*) during this period were M(an)/M = .407, W(oman)/M = .32, M/W = .115, and W/W = .159, the comparable proportions for the articles that these authorship teams cited were M/M = .549, W/M = .257, M/W = .109, and W/W = .085 (Postle and Fulvio, *JoCN*, 34:1, pp. 1–3). Consequently, *JoCN* encourages all authors to consider gender balance explicitly when selecting which articles to cite and gives them the opportunity to report their article's gender citation balance.

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